

Few-cycle pulses focused through annular and conical pupils at high numerical aperture: bandwidth-invariant depth-of-focus extension and the erosion of transverse structure

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Abstract

Structured pupils — annular apertures and conical (axicon) phases — extend the depth of focus (DOF) of a tightly focused beam, while few-cycle pulses carry an octave-scale bandwidth that smears the fine structure of the focal field. What happens when the two are combined at high numerical aperture (NA), where the field is fully vectorial, has not been reported. Using Richards–Wolf vector diffraction with full-spectrum temporal integration, we show that the fractional DOF extension produced by an annular pupil, $\text{DOF}(\varepsilon)/\text{DOF}(0)$, is *invariant to few-cycle bandwidth* — constant to three digits from continuous wave down to a 1-fs pulse ($44.35 \rightarrow 44.33$ at obstruction $\varepsilon = 0.99$, $\text{NA} = 0.8$, $\lambda_c = 750$ nm) and across apertures $\text{NA} = 0.6$ – 1.0 — while the transverse Bessel ring structure erodes *non-monotonically* in pulse width: the Linfoot structural content first dips below unity (moderate pulses wash out the rings) before rising above it (few-cycle broadening of the central lobe). A core/ring decomposition explains the non-monotonicity quantitatively. Supporting results sharpen the deterministic toolkit: scalar-validity thresholds at high NA are shown to be strongly anisotropic (dependent on polarization, observation cut, and direction — a single “NA threshold” is ill-posed); Gaussian apodization *extends* scalar validity; the thin-annulus Bessel core is vector-deformed at high NA while the axicon’s apparent deviation from the Durnin J_0^2 profile is shown to be stationary-phase error, vanishing as $1/\beta$. All results are validated against analytic limits, published anchors, and first-principles Maxwell/Helmholtz residual checks. The bandwidth-invariant DOF extension identifies pulsed annular/Bessel illumination as a robust route to extended- depth nonlinear microscopy with few-cycle sources.

1 Introduction

Tight focusing of ultrashort pulses underlies multiphoton microscopy, micromachining, and strong-field light–matter interaction. At high numerical aperture the focal field must be treated vectorially; the Richards–Wolf (RW) diffraction integrals [1, 2] remain the standard description, and their extension to broadband pulses — summing monochromatic vector fields incoherently over the pulse power spectrum — was analyzed for unstructured (clear circular) pupils by Romallosa, Bantang and Saloma [3] [V], who quantified the smearing of the focal structure of a 750-nm beam at pulse widths down to 1 fs via Linfoot’s image criteria [V].

Independently, *pupil structuring* is a mature route to focal engineering. Annular apertures narrow the central lobe and extend the depth of focus [4, 5]; in the thin-ring limit the focal field approaches Durnin’s nondiffracting Bessel beam [6]; conical (axicon) phases produce an extended focal segment

with a position-dependent Bessel profile [7, 6]. At high NA these structured foci acquire vector character: double-ring and cylindrical-vector illumination produce sub-wavelength needles of controlled polarization over many wavelengths [8, 9] [V]. On the application side, axicon-generated Bessel–Gauss beams extend the two-photon depth of field by an order of magnitude or more [10] [V-mechanism], and Bessel light sheets image deep into scattering tissue [11] [TODO: verify exact numbers before submission].

What is missing between these two literatures is their *combination*: how do structured-pupil foci respond to few-cycle bandwidth at high NA? The bandwidth couples to the pupil geometry nontrivially — every spectral component sees the same geometric pupil but produces a focal pattern whose scale is proportional to its wavelength, so spectral summation washes out wavelength-scale structure (the Romallosa effect [3]) while potentially preserving geometric ratios. To our knowledge no published work reports the pulsed behavior of annular or axicon foci in the vectorial, few-cycle regime (Sec. 4 discusses the closest prior art).

Here we answer that question with a validated deterministic toolkit (Sec. 2) and report four results (Sec. 3): (i) scalar-validity thresholds at high NA are strongly *anisotropic* — they depend on input polarization, on the observation cut, and on direction (transverse vs axial), so a single scalar-breakdown NA is ill-posed; Gaussian apodization systematically *extends* scalar validity; (ii) the annular DOF law $\text{DOF}(\varepsilon)/\text{DOF}(0) = 1/(1 - \varepsilon^2)$ holds to three digits paraxially but falls $\sim 40\%$ short at $\sin \alpha = 0.9$, and the thin-annulus Bessel core is genuinely vector-deformed at high NA (deviation 0.163 from J_0^2 vs 0.004 at low NA); (iii) the axicon’s transverse profile deviation from the predicted local J_0^2 is a stationary-phase artifact that decays as $1/\beta$ (0.134 at $\beta=5$ to 0.006 at $\beta=2560$), not a vector effect — with practical validity ceilings; and (iv) *the headline*: the fractional annular DOF extension is invariant to few-cycle bandwidth across NA = 0.6–1.0, while the transverse ring contrast erodes non-monotonically in pulse width, an effect we explain quantitatively by decomposing the pattern into its Bessel core and rings.

2 Theory and methods

2.1 Vector focal fields

For an aplanatic system of semi-aperture α ($\sin \alpha = \text{NA}/n$) illuminated by an x -polarized beam with pupil amplitude $A(\theta)$, the focal field is given by the RW integrals [2]

$$I_m(r, z; k) = \int_0^\alpha A(\theta) \sqrt{\cos \theta} K_m(\theta) J_m(kr \sin \theta) e^{ikz \cos \theta} d\theta, \quad m = 0, 1, 2, \quad (1)$$

with kernels $K_0 = \sin \theta(1 + \cos \theta)$, $K_1 = \sin^2 \theta$, $K_2 = \sin \theta(1 - \cos \theta)$, and field components $\mathbf{E} \propto (I_0 + I_2 \cos 2\phi, I_2 \sin 2\phi, -2iI_1 \cos \phi)$. Following Ref. [3] we quote the meridional cut with true zeros, $|I_0 - I_2|^2$ (the y - z plane), and verify key results on the orthogonal cut $|I_0 + I_2|^2 + 4|I_1|^2$ and the azimuthal average.

Annular pupils of obstruction ε are treated by field-level Babinet subtraction, $I_m^{\text{ann}} = I_m(\alpha) - I_m(\alpha_{\text{in}})$ with $\sin \alpha_{\text{in}} = \varepsilon \sin \alpha$; apodization uses the truncated-Gaussian $A(\theta) = \exp(-\alpha_t \sin^2 \theta / \sin^2 \alpha)$ of Horváth and Bor [12]; the axicon is the conical pupil phase $A(\theta) = \exp(i\beta \sin \theta / \sin \alpha)$.

2.2 Few-cycle pulses

An unchirped Gaussian pulse of intensity FWHM τ has power spectrum $|E(\omega)|^2 \propto \exp[-(\omega - \omega_c)^2 / 2a]$ with $a = 2 \ln 2 / \tau^2$ (spectral FWHM $4 \ln 2 / \tau$; the transform-limited product $\Delta \nu \Delta t = 2 \ln 2 / \pi$). The

time-integrated focal intensity is the incoherent spectral sum [3]

$$I(\mathbf{r}) = \int_0^{2\omega_c} |E(\omega)|^2 |\mathbf{E}(\mathbf{r}; \omega)|^2 d\omega. \quad (2)$$

The integration window matters: we find (Appendix A) that Ref. [3]’s published anchors are reproduced only when Eq. (2) is evaluated over the *full positive spectral support* $\omega \in (0, 2\omega_c)$; restricting to the FWHM band quoted in the text discards 24% of the spectral weight and fails every anchor.

2.3 Metrics and validation

Pulsed-vs-cw pattern change is quantified by Linfoot’s criteria (fidelity F , structural content S , correlation quality Q ; $2Q - S = F$) on peak-normalized profiles [3, 15]. DOF denotes the axial FWHM of the on-axis intensity in optical units $u = k \sin^2 \alpha z$.

Every layer of the computation is validated before reuse: low-NA limits reproduce the Airy and Durnin J_0^2 patterns to 5×10^{-3} ; Babinet at $\varepsilon=0$ is exact to 10^{-12} ; the published anchors of Ref. [3] (axial $S = 1.058$ vs. 1.057; transverse FWHM 472.8 nm; zero rings at 0.555 and 3.05 μm) and the annular closed forms of Ref. [4] (0.8%) are reproduced. Beyond benchmark matching, the implementation passes *first-principles* residual checks: the scalar field satisfies the Helmholtz equation and the vector field satisfies $\nabla \cdot \mathbf{E} = 0$ at the finite-difference floor ($\sim 10^{-4}$), with deliberately broken controls (wrong k ; E_z kernel factor) failing at the 10^{-1} level — pinning the kernels of Eq. (1) to Maxwell’s equations independently of any literature anchor. A 40-test regression suite locks these results. [\[TODO: cite code/data repository \(Zenodo DOI\) at submission\]](#)

3 Results

3.1 Anisotropic scalar-validity thresholds; apodization extends scalar validity

Comparing vector and scalar predictions of the focal FWHM as NA grows shows that the scalar-breakdown threshold is not a single number (Table 1): at the 1% level the worst case (transverse, linear- x input, cut parallel to the polarization) fails at $\text{NA} \approx 0.30$, while the perpendicular cut holds to ≈ 0.85 and the axial profile to ≈ 1.15 . The anisotropy *is* the longitudinal field: E_z fills the parallel-cut zeros and elongates the spot along the polarization axis. Gaussian underfilling ($\alpha_t=4$) de-weights marginal rays and *extends* the worst-case 1%/5% thresholds from 0.30/0.60 to 0.40/0.85.

Table 1: NA at which the scalar prediction of the focal FWHM deviates by 1%/5% (uniform pupil, $\lambda_c = 750 \text{ nm}$, $n = 1.3$).

quantity	1%	5%
transverse, linear- x , $\parallel$ cut (worst)	0.30	0.60
transverse, circular input	0.45	0.90
transverse, linear- x , $\perp$ cut	0.85	—
axial (any polarization)	1.15	—

3.2 Annular apertures: DOF law and the vector Bessel core

The paraxial law $\text{DOF}(\varepsilon)/\text{DOF}(0) = 1/(1 - \varepsilon^2)$ holds to three digits at low NA (50.13 vs. 50.25 at $\varepsilon = 0.99$) and falls $\sim 40\%$ short at $\sin \alpha = 0.9$ (30.4). The Strehl cost follows $(1 - \varepsilon^2)^2$. In the

thin-ring limit the transverse profile approaches $J_0^2(v)$ with deviation 0.004 at low NA but **0.163** at $\sin \alpha = 0.9$ — a clean, isolated *vector* deformation of the Bessel core (the single-cone geometry isolates the $(1 \pm \cos \theta)$ kernels at one angle).

3.3 The axicon deviation is stationary phase, not vector

The axicon focal segment carries a local Bessel profile $J_0^2(v \sin \theta^* / \sin \alpha)$ at the stationary cone angle $\theta^*(u)$. Its 4–8% deviation from this prediction is *equal* at low and high NA, and a β -scan shows it decays as $1/\beta$ (0.134 at $\beta=5$ to 0.006 at $\beta=2560$): it is the stationary-phase approximation error of the analytic model, not a failure of the field computation (which matches the full Debye integral to 10^{-3} throughout). Practical validity ceilings: the pupil-phase sampling resolves $\beta \gtrsim 2500$, but for a physical objective ($f \approx 2$ mm) the focal segment leaves the Debye focal region at $\beta \approx 160$.

3.4 Headline: pulsed annular pupils — invariant DOF, eroding rings

Figure 1 combines the validated spectral sum [Eq. (2)] with the validated annular fields. Two phenomena coexist:

(a) The DOF extension is bandwidth-invariant. At NA = 0.8, $\text{DOF}(\varepsilon)/\text{DOF}(0)$ changes from 44.35 (cw) to 44.33 (1 fs) at $\varepsilon = 0.99$ — invariance to three digits at a bandwidth where the transverse pattern is strongly smeared. The robustness map (Fig. 2) shows the same invariance across apertures: over $\varepsilon = 0.5$ –0.99 and NA = 0.6–1.0 the cw→1-fs spread of the ratio never exceeds **0.10%** (e.g. $47.30 \rightarrow 47.30$ at NA=0.6, $44.35 \rightarrow 44.33$ at 0.8, $39.16 \rightarrow 39.12$ at 1.0, all at $\varepsilon = 0.99$), with the largest spreads at the highest aperture — consistent with the weak residual NA-dependence of the ratio itself. The mechanism is geometric: each spectral component produces the same annular cone geometry, so its *fractional* DOF extension is wavelength-independent; the spectral sum then averages nearly identical ratios. The absolute DOF of each configuration lengthens only $\sim 1\%$ from cw to 1 fs.

(b) The transverse ring structure erodes non-monotonically. The Linfoot structural content of the transverse pattern (pulsed vs. cw at the same ε) dips *below* unity at moderate pulse widths ($S = 0.988/0.987$ at 5/2 fs, $\varepsilon = 0.9$) before rising above it at 1 fs ($S = 1.058$). A core/ring decomposition at the first cw zero explains this quantitatively: the Bessel-like central lobe is robust (S_{core} rises slowly, $1.00 \rightarrow 1.04$) while the rings first wash out (S_{ring} dips to 0.72 — rings of different spectral components sit at different radii and average destructively) and are then overfilled by the broadened central skirt at 1 fs ($S_{\text{ring}} = 1.37$). The rings carry only 7% of the pattern weight, but with $S_{\text{core}} \approx 1$ the small ring deficit controls the sign of $S - 1$. The effect is robust: it is smooth on a dense τ grid, flat against spectral sampling ($N_{\text{freq}} = 51$ –801), and present in the $\parallel$ cut, the $\perp$ cut, and the full azimuthally averaged vector intensity.

4 Relation to prior work

Ref. [3] treats few-cycle tight focusing for the clear pupil; the annular/Bessel DOF literature [4, 6, 10] is monochromatic or narrowband; high-NA vector needle work [8, 9] is cw. Broadband extensions of Debye focusing theory exist [13] [V] but do not address structured pupils’ pulsed response. A recent study treats a focused pulsed beam in a scattering medium [14] but for the clear pupil and without the DOF metric. Targeted literature searches (“pulsed annular depth of focus,” “few-cycle Bessel focus,” “bandwidth-invariant depth of focus”) return no prior report of the bandwidth-invariance of the annular DOF extension or of the non-monotonic ring erosion.

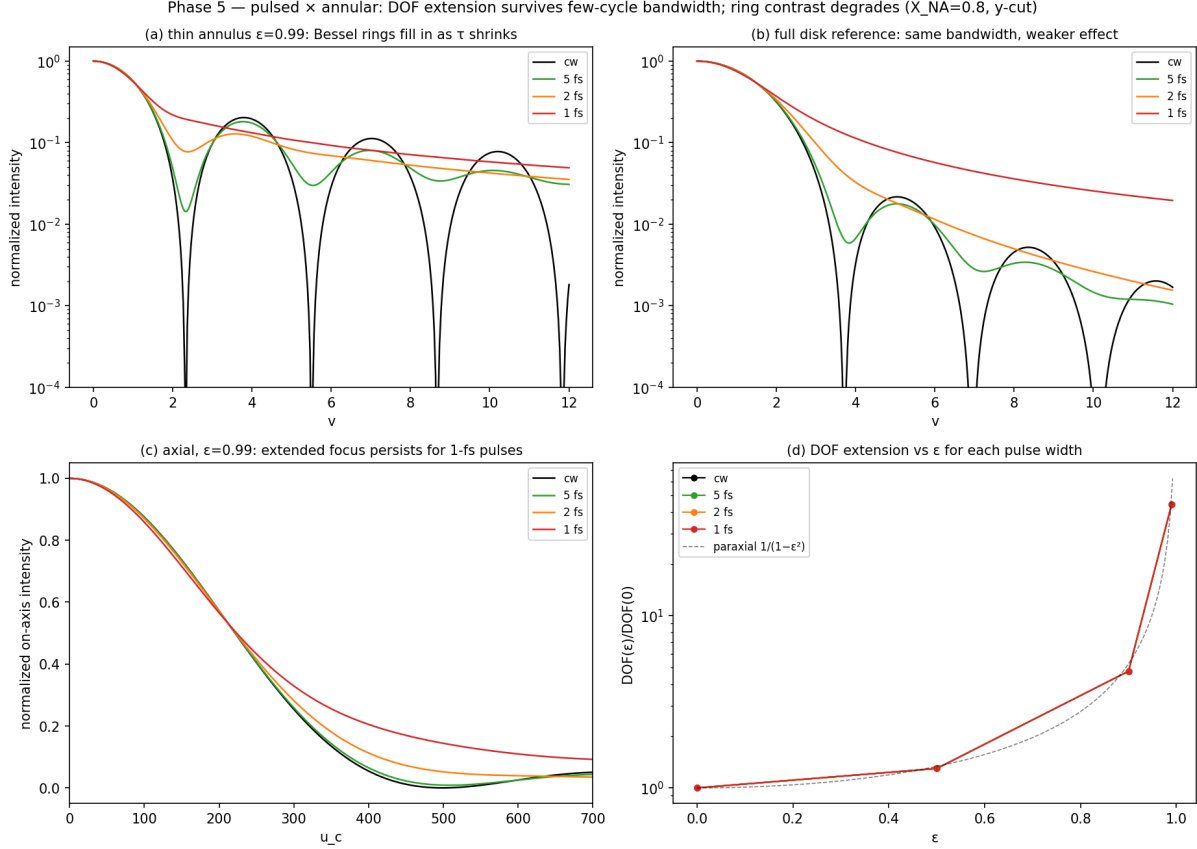


Figure 1: Pulsed annular focusing at $NA = 0.8$: transverse ring filling vs. τ at $\epsilon = 0.99$; full-disk reference; persistence of the extended axial focus at 1 fs; and the DOF ratio following the paraxial law for all τ . [TODO: replace with publication-quality re-render]

5 Discussion and outlook

For applications, the message is enabling: the axial benefit of annular/Bessel illumination survives the broadest usable bandwidths, so extended-DOF nonlinear microscopy [10] can employ few-cycle sources without sacrificing the depth advantage — what is traded is transverse ring contrast, and in two-photon operation the rings are already suppressed by the intensity-squared response. The anisotropic thresholds (Sec. 3.1) give practitioners a lookup for when scalar modeling suffices, per polarization and per observable. The validated deterministic fields constructed here are the source terms for a companion study of these structured pulsed foci propagating through scattering media (in preparation), where the wavelength dependence of scattering couples the spectral and spatial degrees of freedom.

A The spectral integration window of Ref. [Romallosa 2003]

Integrating Eq. (2) over the FWHM band (483 nm–1.67 μm as quoted in the text of Ref. [3]) discards 24% of the spectral weight at $\tau = 1$ fs and yields transverse $S = 1.020$, inconsistent with the published 1.062. Integrating the full positive support $\omega \in (0, 2\omega_c)$ reproduces every anchor (axial $S = 1.058$ vs. published 1.057; $F \approx 0.96$; the pulsed pedestal ≈ 0.05 at $r = 1 \mu\text{m}$ of their Fig. 2). We conclude the

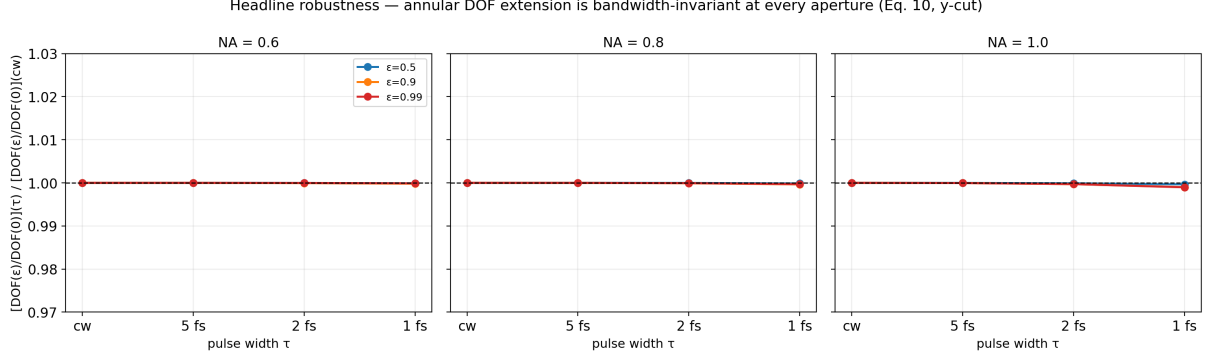


Figure 2: Robustness of the headline: the normalized DOF ratio $[\text{DOF}(\varepsilon)/\text{DOF}(0)](\tau)$ relative to its cw value, for $\varepsilon = 0.5, 0.9, 0.99$ at $\text{NA} = 0.6, 0.8, 1.0$.

published band describes the FWHM, not the integration window, and recommend the full window for reproducibility.

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